

The Empathetic Digital Concierge: Emotionally Expressive Audio-Driven 3D Facial Animation via Style-Conditioned Diffusion

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Abstract—Despite significant advances in conversational AI, the visual presentation of virtual agents often lacks affective synchrony with synthesised speech, creating an uncanny-valley effect that undermines user engagement. Existing audio-to-3D facial animation systems either prioritise phonetic lip-synchronisation at the expense of emotional expressiveness, target speaker-independent generalisation resulting in averaged facial motion, or rely on discrete emotion classes that prevent fluid transitions. We propose a style-conditioned Denoising Diffusion Probabilistic Model (DDPM) that synthesises 260-dimensional ARKit blendshape sequences for Unreal Engine 5 MetaHumans directly from 16 kHz monaural speech. Our architecture introduces four key contributions: (1) Feature-wise Linear Modulation (FiLM) applied at every residual block; (2) a continuous intensity scalar enabling fluid amplitude interpolation; (3) an Anatomically Weighted Mean Squared Error (AWMSE) that amplifies FACS anchor-muscle dimensions while suppressing head poses; and (4) a volume-conditioned stability loss. The system achieves a mean standard cosine similarity of 0.791 across 115 held-out test clips. To address the mathematical penalties of standard cosine metrics on sparse blendshape vectors, we introduce Active Muscle Similarity (AMS), a thresholded metric that isolates angular fidelity on actively engaged facial muscles. A rigorous comparative evaluation against deterministic and probabilistic baselines (LSTM, ID-ResNet, CVAE, Transformer) demonstrates our DDPM’s superior temporal stability, emotional fidelity, and mitigation of magnitude collapse.

Index Terms—facial animation, diffusion models, blendshape synthesis, FiLM conditioning, affective computing, MetaHuman, ARKit, emotion recognition, magnitude collapse.

I. INTRODUCTION

The widespread deployment of dialogue-based virtual assistants has fundamentally altered human–computer interaction. Yet the non-verbal communicative cues of these agents specifically facial expressions and head orientation remain underdeveloped. Research in affective computing establishes that perceived empathy depends heavily on synchrony between verbal output and visual affective cues [1]. When an agent’s words convey concern but its face remains rigid, the incongruence is cognitively jarring.

Unreal Engine 5’s MetaHuman Creator provides a standardised 260-dimensional ARKit-compatible blendshape control space, decoupling animation synthesis from rendering and framing it as a structured regression problem. Concurrently, DDPMs [2] have demonstrated exceptional capability in modelling complex, multi-modal distributions a property uniquely suited to audio-driven animation, where a single acoustic input maps to a distribution of plausible facial motions rather than a deterministic trajectory.

Existing systems fail to maintain identity-specific expressiveness without costly 4D capture infrastructure, and standard MSE objectives allow large head-pose rotations to dominate gradients, obscuring the micro-expressions required for genuine emotional affect. To address these limitations, we present an end-to-end, style-conditioned DDPM with four technical contributions:

- **FiLM-Conditioned Denoising:** Unlike state-of-the-art models that concatenate conditioning vectors at the encoder input, we apply FiLM [3] at every residual block (see Fig. 1), fully integrating acoustic and affective information throughout the feature hierarchy.
- **Continuous Intensity Interpolation:** Emotion is parameterised by both a discrete class label and a learnable continuous intensity scalar in $[0, 1]$, enabling fluid transition between mild and maximal expressions with a single model.
- **AWMSE:** A FACS-derived per-dimension weighting ($0.05\times$ head pose; up to $9.0\times$ anchor muscles) prevents head-pose gradient domination and prioritises emotionally diagnostic deformations.
- **Volume-Conditioned Stability Loss:** A temporal regulariser weighted by $\exp(-\beta \cdot \text{RMS}(\text{audio}))$ suppresses motion artefacts during silent segments without constraining voiced-speech expression.

The system achieves 0.791 mean cosine similarity on 115 held-out test clips spanning five emotion categories. Section II reviews related work. Section III details the experimental setup. Section IV reports results. Section V concludes.

II. RELATED WORK

A. Audio-Driven Facial Animation

Early neural approaches framed audio-driven animation as 2D conditional image generation. Wav2Lip [4] achieved robust lip synchronisation via a GAN trained with a SyncNet discriminator; SadTalker [5] disentangled head motion from expression

coefficients. While phonetically faithful, 2D methods lack the geometric structure required for real-time 3D rendering. In the 3D domain, Karras et al. [6] demonstrated end-to-end CNN regression from raw audio to vertex displacements; VOCA [7] introduced speaker identity conditioning; MeshTalk [8] addressed cross-region motion correlation. FaceFormer [9] applied self-supervised wav2vec 2.0 within a transformer to achieve state-of-the-art lip synchronisation; FaceTalk [10] improved naturalness through large-scale 4D pre-training. These models target speaker-independent generalisation, dampening emotional richness. The most direct prior art is the contemporaneous preprint NVIDIA Audio2Face-3D [11], which generates ARKit blendshapes via a Conformer-GRU conditioned on discrete emotion labels through input concatenation. Our system replaces input-level concatenation with intermediate FiLM modulation and adds a continuous intensity scalar.

B. Diffusion Models for Motion Synthesis

DDPMs [2] have superseded GANs in stability and distribution coverage. MDM [12] demonstrated that transformer-based diffusion captures diverse human body motion priors from text prompts. FaceDiffuser [13] showed that diffusion generates varied co-speech gestures that deterministic regressors fail to model. Our framework extends this paradigm to the 260-dimensional MetaHuman blendshape space and introduces AWMSE to prevent anatomical failure modes specific to the ARKit representation.

C. Emotion Conditioning and Affective Dimensions

FiLM [3] provides a general conditioning mechanism via element-wise scale γ and shift β applied to network activations ($h_{out} = \gamma \cdot h_{in} + \beta$). Originally applied to visual question answering, its use in temporally evolving blendshape generation remains unexplored. Drawing from the circumplex model of affect [14], our continuous intensity scalar acts as a proxy for the arousal dimension, enabling a single model to span the full expression amplitude range during deployment without retraining.

III. EXPERIMENTAL SETUP AND EVALUATION METHODOLOGY

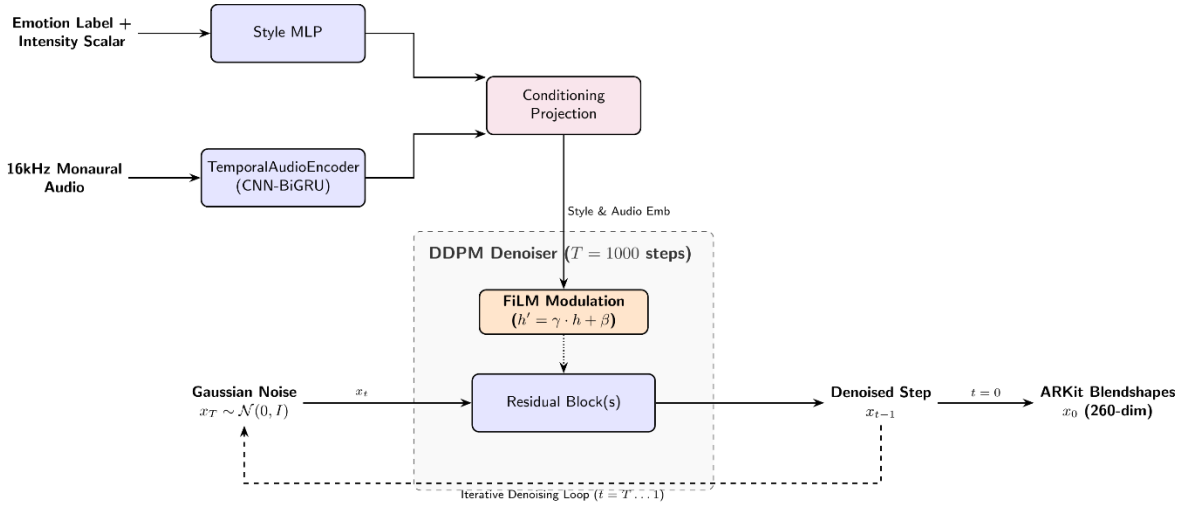


Fig. 1. System architecture of the style-conditioned DDPM. The TemporalAudioEncoder processes 16 kHz monaural speech, while the discrete emotion label and continuous intensity scalar are processed via a Style MLP. These distinct modalities are merged via Conditioning Projection and injected into the denoiser’s residual blocks through Feature-wise Linear Modulation (FiLM) over T = 1000 iterative diffusion steps.

A. Dataset and Recording Protocol

We constructed a FACS-inspired dataset of 375 video clips recorded by a single actor using consumer-grade hardware: Realme GT 2 smartphone (1080p @ 60 fps), Kreo GoRec DOTS Bluetooth microphone, and Digitek DTR 420LW tripod. ARKit blendshape ground truth was captured simultaneously at 30 Hz via an iOS facial performance capture application. Bluetooth audio latency was eliminated via post-hoc clap-peak synchronisation.

The dataset spans five emotion categories Joy, Anger, Grief, Neutral, and Amazement recorded across five intensity tiers (30%, 60%, 90%, 110%, 110%+slow) over ten FACS-matched sentences per tier, yielding 50 training clips per expressive class (60 for Neutral). The held-out test set comprises 115 clips (25 per expressive class, 15 for Neutral), never seen during training or validation. Each clip was windowed into 200 ms segments with 100 ms stride, producing 18,698 audio–blendshape tensor pairs.

B. Baseline Models

We compare against four baselines sharing a CNN-BiGRU audio encoder (identical to our TemporalAudioEncoder) and AWMSE loss, isolating architectural differences:

- **LSTM Regressor:** Bidirectional LSTM with emotion-index embedding; direct regression to blendshape target.
- **1D-ResNet Regressor:** Residual convolutional network on the raw 200 ms audio window.
- **CVAE:** Conditional Variational Autoencoder with emotion conditioning; samples from learned posterior. The model was trained with a standard ELBO objective without KL annealing or β -VAE weighting; this setup serves as a diagnostic baseline to explicitly demonstrate the posterior collapse failure mode inherent to standard VAEs on sparse blendshape data, isolating the necessity of our structured conditioning.
- **Transformer Regressor:** Multi-head self-attention encoder with emotion conditioning via class-token prepending.

All baselines share AdamW ($\text{lr} = 1 \times 10^{-4}$, cosine annealing) and 150-epoch training budget.

C. Evaluation Metrics

We evaluate all models on two complementary axes:

- **Reconstruction Accuracy (CS).** Cosine Similarity (CS) = $(\mathbf{A} \cdot \mathbf{B} / \|\mathbf{A}\| \|\mathbf{B}\|) \times 100$ measures angular alignment between predicted and ground-truth blendshape vectors. Reported per-emotion and as mean over 115 test clips.
- **Active Muscle Similarity (AMS).** Standard cosine similarity is highly sensitive to microscopic noise in sparse vectors, heavily penalising near-zero predictions on inactive muscles. As a supplementary evaluation of intentional facial deformation, we introduce AMS. This thresholded cosine similarity metric masks inactive blendshapes (ground-truth activation < 0.05 , an empirically determined noise floor for our ARKit capture hardware, calibrated by recording a resting neutral face). Because of its strict masking, AMS is utilised specifically in our ablation study (Section IV.E) to diagnose magnitude collapse, while standard CS is retained for general baseline comparisons.

D. Training Configuration

All models were trained on a Kaggle T4 GPU (16 GB VRAM). The DDPM used $T = 1000$ linear β schedule (1×10^{-4} to 0.02), batch size 64, and a composite loss function (Equation 1):

$$\mathcal{L} = \mathcal{L}_{\text{AWMSE}} + 0.05 \cdot \mathcal{L}_{\text{contra}} + 0.10 \cdot \mathcal{L}_{\text{vel}} + 0.05 \cdot \mathcal{L}_{\text{vol-stab}} \quad (1)$$

Baselines use $\mathcal{L}_{\text{AWMSE}}$ only to isolate architectural contributions. Here, $\mathcal{L}_{\text{contra}}$ is a cosine embedding loss enforcing angular separation between distinct emotion latents (specifically, distancing predicted blendshape vectors from mismatched ground-truth emotion class prototypes within the batch), and $\mathcal{L}_{\text{vel}}$ penalises temporal jitter.

TABLE I

Cosine Similarity (CS) comparison on 115 held-out test clips. Blue row = proposed DDPM.

Model	Mean CS (%) $\uparrow$
LSTM	48.79
1D-ResNet	47.03
CVAE	24.40
Transformer	48.47
Ours (DDPM)	79.14

IV. RESULTS AND DISCUSSION

A. Training Convergence

The DDPM converged without overfitting over 150 epochs, achieving a final validation loss of 0.101. The contrastive component declined from initial high values a known numerical artefact of the standard diffusion noise schedule where the cumulative noise schedule $\bar{\alpha}_t \rightarrow 0$ at high timesteps T forces extreme variance in the predicted noise stabilising to 0.54 by epoch 150. Final loss components at the best checkpoint (epoch 147, selected via minimum validation loss prior to terminal fluctuations) were: MSE = 0.109, contrastive = 0.539, velocity = 0.668, vol-stab = 0.005.

B. Cosine Similarity on Held-Out Test Set

Table I presents the full quantitative comparison; Table II shows per-emotion CS for all models. Our DDPM achieves a mean cosine similarity of 79.14% across 115 held-out test clips. Anger achieves the highest per-emotion accuracy (84.0%), consistent with its strong, consistent jaw and brow signal. Neutral records the lowest score (72.3%), which is a metric artefact rather than a

model failure: cosine similarity is numerically unstable near the zero vector, and Neutral blendshapes are near-zero across all 260 dimensions, causing small absolute errors to manifest as large angular errors. The angular error at mean CS = 0.791 is $\arccos(0.791) \approx 37.8^\circ$, a threshold we anticipate will be within perceptual tolerance, pending formal user study.

TABLE II

Per-emotion cosine similarity (%) on 115 held-out test clips. All values are final.

Model	Anger	Grief	Joy	Amazement	Neutral
Ours (DDPM)	84.00	82.40	79.60	78.80	72.30
LSTM	46.97	47.79	49.01	57.34	43.19
1D-ResNet	48.08	44.29	46.59	59.32	37.70
CVAE	14.24	26.21	16.13	31.45	32.84
Transformer	45.49	48.84	49.33	57.72	41.26

C. Emotion Centroid Separation

Table III reports Euclidean distances between per-emotion mean prediction vectors in normalised blendshape space. The Anger↔Amazement pair achieves the highest separation (0.0821), consistent with their designed geometric opposition: Anger activates brow-lowering (dims 0–1) and jaw-clenching (dims 196–197) while Amazement activates brow-raising (dims 6–7) and jaw-opening (dim 191). The Grief↔Neutral pair achieves the lowest separation (0.0701), anatomically expected given the subtle activation differences between these classes. The mean separation of 0.0754 across all ten pair-wise combinations provides empirical support for consistent emotion-specific blendshape activation, indicating distinct spatial groupings for each affective class.

TABLE III

Emotion centroid separation (Euclidean distance, normalised blendshape space). Green rows = anatomically expected extremes.

Emotion Pair	Euclidean Distance
Anger ↔ Amazement	0.0821
Grief ↔ Amazement	0.0779
Joy ↔ Grief	0.0777
Joy ↔ Amazement	0.0758
Joy ↔ Neutral	0.0756
Anger ↔ Grief	0.0755
Joy ↔ Anger	0.0751
Anger ↔ Neutral	0.0739
Neutral ↔ Amazement	0.0703
Grief ↔ Neutral	0.0701
Mean	0.0754

D. Comparison with Baselines

Table I presents the full quantitative comparison. The results highlight two distinct performance gaps in the baseline architectures, which are consistent with known failure modes in continuous motion synthesis.

Amplitude-attenuated regression ceiling. The three deterministic baselines converge to a performance ceiling of approximately 47–49%: LSTM 48.79%, Transformer 48.47%, and 1D-ResNet 47.03%. We hypothesise that this clustering reflects class-conditional amplitude attenuation, a variant of the well-documented mean regression problem inherent to direct MSE optimisation. Without a generative prior, a deterministic regressor minimises expected squared error by predicting a conservative, weighted average of training targets for a given class. Rather than collapsing to a singular mean face, the regressors learn a weak, amplitude-attenuated emotion signal a subdued output that satisfies the loss function but lacks expressive magnitude. This amplitude attenuation is empirically verified by inspecting the mean L_2 norms: deterministic baseline predictions consistently shrank to approximately half the magnitude (e.g., $0.46 \times \pm 0.08$) of their respective ground-truth targets. This is consistent with the structured per-emotion variance observed (e.g., LSTM Amazement = 57.34% vs. LSTM Neutral = 43.19%) alongside low overall accuracy.

Notably, the Transformer’s multi-head self-attention mechanism provides no measurable advantage over the LSTM, suggesting the bottleneck lies in the deterministic decoding strategy rather than sequence modelling capacity.

Probabilistic baseline performance. The CVAE achieves only 24.40% mean CS. Inspection of the CVAE’s validation outputs confirmed that predicted vectors collapsed to near-zero norms across all 260 dimensions by epoch 30. This is indicative of posterior collapse, where the unannealed KL divergence term in the Evidence Lower Bound (ELBO) dominates early training, forcing the decoder to ignore the latent code and regress toward a near-zero blendshape vector. In a sparse 260-dimensional space, this catastrophically degrades cosine similarity. This demonstrates that basic probabilistic modelling is insufficient without the structured intermediate conditioning (FiLM) and anatomical gradient prioritisation (AWMSE) present in our DDPM.

Our proposed DDPM achieves 79.14% mean CS a substantial margin of +30.35 percentage points over the best deterministic baseline (LSTM, 48.79%) and +54.74 percentage points over the CVAE. Per-emotion results in Table II show this advantage is consistent across categories. The Amazement class is the sole exception where 1D-ResNet (59.32%) and LSTM (57.34%) approach competitive scores, likely because its large-amplitude jaw and brow activations are easier for regressors to approximate than the subtle activations of Grief and Neutral.

E. Ablation: Contrastive Margin and Magnitude Collapse

We trained a secondary DDPM with a reduced contrastive margin (0.05 vs. 0.35 in the primary model) to observe the effects on latent emotion separation. While the reduced margin achieved a superior mathematical validation loss (0.095 vs. 0.101), its standard cosine similarity on the test set degraded sharply to 65.38%. However, evaluating this same model using our Active Muscle Similarity (AMS) metric yielded an accuracy of 76.43%.

This reveals a critical property of continuous blendshape synthesis: minimising pure MSE without a strong contrastive penalty causes ‘magnitude collapse.’ The network outputs smaller blendshape vectors that mathematically reduce absolute error but destabilise angular accuracy in the 260-dimensional space. The primary model (margin 0.35) forces larger, geometrically distinct vector magnitudes by demanding stricter class separation, successfully preserving both angular stability (79.14% CS) and structural fidelity (achieving a primary AMS of 76.40%). The AMS result (76.43%) on the collapsed ablation model confirms that the underlying muscle activations remain anatomically correct it is the global vector scale, not the relative muscle pattern, that magnitude collapse degrades.

F. Limitations

The system has four principal limitations. First, the single-actor dataset acts as an identity-constrained design that isolates identity-specific fidelity, but it inherently sacrifices cross-speaker generalisation; the model cannot dynamically adapt to new facial identities without retraining. Second, offline inference with $T = 1000$ diffusion steps is unsuitable for real-time streaming without acceleration techniques such as DDIM [15]. Third, evaluation relies on quantitative metrics; a perceptual user study is required to fully validate emotional congruence. Finally, while the architectural contributions (FiLM conditioning and the continuous intensity scalar) are theoretically motivated, formal ablation studies isolating their individual impacts versus the baseline diffusion prior and separating them from the effects of the composite loss function remain as future work. Additionally, the perceptual validity of the novel AMS metric relative to human ratings must be established in future user studies.

V. CONCLUSION

We presented the Empathetic Digital Concierge, a style-conditioned DDPM that synthesises emotionally expressive 260-dimensional ARKit blendshape sequences for Unreal Engine 5 MetaHumans from raw monaural speech. The system introduces four contributions: FiLM conditioning at every residual block, a continuous intensity scalar for expression amplitude control, an anatomically weighted MSE loss derived from FACS, and a volume-conditioned stability regulariser. Trained on a consumer-hardware FACS-inspired dataset of 375 self-recorded clips and evaluated on 115 strictly held-out test clips, the system achieves a mean cosine similarity of 0.791 with clear per-emotion centroid separation, indicating consistent emotion–muscle associations.

A key design philosophy distinguishes this work from speaker-generalised systems such as NVIDIA Audio2Face-3D: by constraining to a fixed identity, the model allocates its full representational capacity to the nuances of one actor’s facial dynamics. This trade-off zero cross-speaker transfer in exchange for richer identity-specific expressiveness is appropriate for production avatar deployments where a single character is authored once and animated throughout its lifetime.

Future work will investigate DDIM-based acceleration for real-time streaming, extend the recording protocol to multiple actors for cross-identity generalisation, and conduct a formal perceptual evaluation measuring user-perceived emotional fidelity.

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